ASSIGNMENT #3: Anonymization of a Dataset with Risk and Utility Analysis

Due date: May 31, 23:59

Important notice: Assignment will be discussed (mandatory) in the following lab class (June 1st or 2nd, depending on your class schedule).

Grading: Assignment #4 is worth **2 points**

TO BE DONE IN **GROUPS OF THREE (MANDATORY)**

The goal of this assignment is to perform the anonymization of a larger dataset. You will resort to the adult dataset, which is a common choice for evaluating anonymization models: https://archive.ics.uci.edu/ml/datasets/adult.

1. Create a new ARX project and import the adult dataset.
2. Analyze the attributes and classify them into Identifying, QIDs, Sensitive or Insensitive. Justify your choice.

   This should take into consideration the values for distinction and separation.
3. You should now characterize/analyze the privacy risks of the dataset in original form.

   Re-identification risk of the original Dataset
4. You should **apply two anonymization models** to your dataset and conduct an analysis of the performance of each model applied to the dataset.

   For the analysis you should consider:

   - The re-identification risk of the anonymized dataset vs the original dataset
   - The utility level, measured through appropriate utility metrics and/or analysis of data distributions
   - Assess the effect on re-identification risk and utility metrics of various configuration parameters of the privacy models (e.g. value of k, suppression limit, coding model, etc), producing suitable plots with the risk and utility as a function of the parameters tested.
5. If the results are not satisfying, you should perform extra iterations, by either adapting the parameters of the privacy models (e.g. suppression limit, coding model, attribute weights, etc) or considering other privacy models to apply.

Submit in moodle (1) your ARX project with anonymization models applied and a (2) document (max 5 pages) explaining (2-a) the classification of attributes, and (2-b) the above risk and utility plots and corresponding analysis.

Evaluation Criteria

- Classification of attributes and justification [15%]
- Coding Models [20%]
  - Definition and configuration of coding model (e.g. hierarchies)
- Privacy models: utility and risk assessment [50%]
  - Analysis of the utility and privacy (re-identification risk) levels (before and after anonymization)
  - Analysis of privacy models' results according to varying privacy model's parameters (e.g. different k values, suppression limit, generalization vs suppression, etc)
- Structure and organization of the report [15%]
  - Documentation of the use of AI (when applicable)